

A Ceiling Dark Matter Cannot Impose: the Bounded-Boost Theorem for MOND-Class Kernels, and What It Says About Galaxies and Clusters

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v4: the kernel swap the theorem selects is carried through the full gate ladder

Abstract

Any modified-gravity kernel [1, 2, 3, 4] that writes the observed acceleration as $g = \nu(y) g_N$, $y = g_N/a_0$, with a Newtonian limit $\nu \rightarrow 1$ and a deep-MOND limit $\nu \rightarrow y^{-1/2}$, makes a prediction that a dark-matter halo structurally cannot make: the *acceleration excess* $\Delta \equiv g_{\text{obs}} - g_{\text{bar}}$ is bounded above by a pure number times a_0 . For the exponential carrier $g_{\text{bar}} = g(1 - e^{-g/a_0})$ the bound is exact and elementary, $\Delta \leq a_0/e = 0.367879\dots a_0$, attained at $g_{\text{bar}} = (1 - 1/e)a_0$; the completed kernel saturates there, so the bound is attained on a plateau. Across five standard kernels the constant lies in $[0.250, 1.000]$, so the ceiling is kernel-independent up to an $O(1)$ factor. In a halo model the same quantity is g_{halo} , set by M_{200} and concentration, which span decades and are not tied to a_0 ; the ceiling is therefore a one-sided hard wall with no fitting freedom, and a single system above it falsifies the kernel. We test it. (i) On 2352 SPARC [5] rotation-curve points beyond 2 kpc from 144 galaxies (the survey’s own $Q < 3$, $i > 30^\circ$ cuts), 99.23% obey the widest kernel bound at both a_0 footings; the eighteen $> 3\sigma$ exceptions lie in five named galaxies, thirteen of them in NGC 5985 alone. (ii) The ceiling is tight enough to discriminate between kernels: 9.8% of those points (263, spread over 32 of 144 galaxies) exceed the exponential carrier’s own ceiling a_0/e at $> 3\sigma$, and between $g_{\text{bar}} = 1$ and $4.5 a_0$ the measured excess is 2–3 $\times$ that ceiling against 1.2–1.9 $\times$ the wider ν_{RAR} [6] ceiling. The coherent stellar-normalisation freedom is independently pinned to +1.8% by the Newtonian limit, and raising a_0 cannot absorb the discrepancy (it would need 12 $\times$ the canonical value). (iii) In X-COP [8, 9] cluster cores, after correcting a radius-unit error we verify directly from the FITS headers, the ceiling fails by a factor 9.1 against the exponential carrier’s bound and 3.4 against the widest bound of any kernel, so no choice of interpolation function can absorb it. Both escapes close quantitatively: nonthermal support would need $\sigma_{\text{1D}} = 673\text{--}1155 \text{ km s}^{-1}$ against the Hitomi Perseus measurement of 164 ± 10 [10], and unseen baryons would need 3.9–5.2 $\times$ the directly imaged X-ray gas. Clusters therefore require a genuine extra source *as a theorem rather than a fit*. The required source is baryon-tracing, but so is the published NFW [7] fit, so its profile shape does not discriminate; we report that as a null. We then carry the swap the theorem selects through the framework’s whole constraint ladder. The kernel enters every gate through a single function, $J_Y(s) = s/\Delta(s)$, whose saturated slope is e for the exponential carrier and $1/C_{\text{RAR}} = 1.544$ for ν_{RAR} : one ratio, 1.76, is the entire content of the swap. It is admissible everywhere. Post-Newtonian α_1 is untouched because the screening crushes its scalar drag to under 2% at either kernel’s own floor; the cosmological linear-source screening is untouched because, re-derived from the action, it contains no kernel dependence at all; and the dark sector’s deep-MOND length is *identical* for the two kernels, only its saturated value differing, by 0.568. Clusters, as the theorem requires, are not rescued: the excess is 5.2 $\times$ ν_{RAR} ’s wider ceiling against 9.1 $\times$ the carrier’s. Finally, we close the loop the corrections open: the programme’s own wide-binary prediction was evaluated at floors the carried kernel does not

permit and with an interpolation that is wrong above the saturation point, and we recompute it. The pre-registered estimator returns $\gamma_v = 1.0375$ and 1.0275 for the exponential carrier at its corrected floors and 1.0450 and 1.0300 for ν_{RAR} carried, against a stale $1.032/1.040$. The two corrections act in *opposite* directions and are of comparable size, so the net change is not monotone in the coherence length and had to be computed rather than assumed. Every value remains far below the pre-registered band, so the prediction stays sharply distinguishable, while the anisotropy sign-flip radius moves out of the survey's usable separation range. This is a preprint of an AI-assisted draft. It claims no completed relativistic theory, no derivation of a_0 or of $\kappa = \frac{1}{2}$, and no empirical detection.

1 The action and the kernel

The candidate under test, in the lineage of relativistic MOND completions [13], is general relativity plus a clock scalar τ with unit normal $n_\mu = -\partial_\mu \tau / \sqrt{-(\partial\tau)^2}$ and projector $q_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$, and one MOND scalar ϕ with $Q = n^\mu \partial_\mu \phi$, $V_\mu = q_\mu^\nu \partial_\nu \phi$, $Y = V \cdot V$:

$$S = \int d^4x \sqrt{-g} \left\{ \frac{1}{16\pi G} [R - 2\Lambda] - c_1 (\nabla_\mu n_\nu) (\nabla^\mu n^\nu) - c_2 (\nabla \cdot n)^2 - c_3 (\nabla_\mu n_\nu) (\nabla^\nu n^\mu) + c_4 (n \cdot \nabla n)^2 \right. \\ \left. + 2(2 - K_B) J^\mu \partial_\mu \phi - (2 - K_B) \mathcal{J} \left(Y + \xi^2 q^{\lambda\sigma} q^{\mu\nu} \nabla_\lambda V_\mu \nabla_\sigma V_\nu \right) - K(Q) \right\} + S_m[g, \psi_m], \quad (1)$$

with $J^\mu = n^\nu \nabla_\nu n^\mu$ the clock's four-acceleration, $K(Q) = K_2(Q - Q_0)^2$, $c_1 = -c_3 = K_B$ so that $c_{13} = 0$ and $c_T = c$, and matter minimally coupled. Nothing below depends on this particular action: the theorem applies to any kernel of the stated class. We carry both acceleration footings, $a_0 = 9.3619 \times 10^{-11}$ and $1.1279 \times 10^{-10} \text{ m s}^{-2}$, throughout; $\kappa = \frac{1}{2}$ in $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ is fitted, not derived.

In the static weak-field limit the scalar carries an exponential kernel. Writing $y = g/a_0$ for the total field, the matter-sourced carrier obeys $g_{\text{bar}} = g(1 - e^{-y})$ for $y \leq 1$, and above $y = 1$ the scalar force saturates at a_0/e .

2 The theorem

Define the *acceleration excess* in units of a_0 ,

$$\Delta(g_{\text{bar}}) \equiv \frac{g_{\text{obs}} - g_{\text{bar}}}{a_0} = y[\nu(y) - 1]. \quad (2)$$

For the exponential carrier this is exactly

$$\Delta = y e^{-y}, \quad \frac{d\Delta}{dy} = (1 - y)e^{-y}, \quad \Delta_{\text{max}} = \frac{1}{e} = 0.367879 \dots \text{ at } y = 1, \quad (3)$$

verified symbolically: the critical point is unique, the second derivative there is $-e^{-1} < 0$, and both endpoint limits vanish. The maximum is reached at $g_{\text{bar}} = (1 - 1/e)a_0 = 0.6321a_0$. Because the completed kernel saturates at $y = 1$, the excess does not fall back but stays at a_0/e : the bound is attained on a plateau.

The statement is not special to this kernel. Δ vanishes as $\sqrt{y}$ at small y and tends to 0 or a finite constant at large y whenever $\nu \rightarrow 1$ at least as fast as $1/y$; being continuous in between, it is bounded. Table 1 gives the supremum for five standard kernels, computed on a 2×10^6 -point logarithmic grid.

kernel	$C = \sup \Delta$	at g_{bar}/a_0
deep-MOND $\sqrt{\cdot}$, $g = \sqrt{a_0 g_N}$	0.2500	0.250
standard μ	0.3003	0.486
exponential carrier (this action)	0.3679	≥ 0.632 (plateau)
ν_{RAR} (McGaugh et al. 2016)	0.6476	2.54
simple μ	1.0000	$y \rightarrow \infty$

Table 1: The ceiling is kernel-independent up to an $O(1)$ constant.

3 Why a dark-matter halo cannot make this prediction

In a halo model the same observable is $g_{\text{obs}} - g_{\text{bar}} = g_{\text{halo}}$, the halo’s own acceleration at that radius. It is set by the halo mass and concentration, which vary by orders of magnitude across the galaxy population and have no relation to a_0 . There is no mechanism placing a common upper bound on g_{halo} , still less one at a fixed multiple of an acceleration defined by the cosmological constant. A ceiling holding across every system and every radius is therefore a coincidence a halo model must absorb as a fine-tuning of the mass–concentration relation against the baryon distribution.

Two properties make this sharper than the radial-acceleration relation from which it descends. It is *one-sided*: only excursions above the wall count, so scatter, distance errors and inclination errors cannot manufacture a false confirmation, only a false violation. And it is *parameter-free* once a_0 is fixed: there is no amplitude, slope or scatter to fit, so a single system above the wall falsifies the kernel outright.

4 Test on rotation curves

We use the SPARC Rotmod curves [5] with the survey’s published sample cuts ($Q < 3$, $i > 30^\circ$) and its fixed $3.6 \mu\text{m}$ mass-to-light ratios, $\Upsilon_{\text{disk}} = 0.5$, $\Upsilon_{\text{bul}} = 0.7$, keeping points with $\delta V/V < 0.10$: 144 galaxies, 2681 points, of which 2352 lie beyond 2 kpc. The uncertainty on Δ combines the velocity error with a 0.1 dex stellar-population spread.

Against the *widest* kernel bound $C = 1.000$, 99.23% of points beyond 2 kpc comply at both footings. The eighteen $> 3\sigma$ exceptions lie in five galaxies: ESO 563-G021, NGC 5907, NGC 5985, NGC 6674 and UGC 12506. Thirteen are in NGC 5985 alone, which has $Q = 1$ and $i = 60^\circ$ and so passes every standard cut; its $V_{\text{flat}} = 294 \text{ km s}^{-1}$ against a baryonic Tully–Fisher expectation near 195 km s^{-1} makes it a genuine exception, and we keep and name it rather than cutting it. Three of the remaining four are near edge-on ($i \geq 83^\circ$) and contribute one or two points each.

The ceiling is tight enough to discriminate between kernels, and it does not favour the kernel this action carries. Table 2 compares the binned median excess with the exponential carrier’s saturated ceiling and with ν_{RAR} .

Beyond 2 kpc, 9.8% of points (263 of 2352, spread over 32 of 144 galaxies) exceed the exponential carrier’s own ceiling a_0/e by more than 3σ . That is the ordinary population, not a tail. Two escapes are closed. A coherent shift of the stellar normalisation is pinned independently by the Newtonian limit, which every kernel shares: at $g_{\text{bar}} > 10 a_0$, where all kernels require $\Delta \rightarrow 0$, the measured median of $g_{\text{obs}}/g_{\text{bar}}$ over 90 points is 1.018, so the normalisation is already correct to 1.8%. Raising a_0 does not help either: the ceiling is a_0/e in physical units, and accommodating the largest measured

g_{bar}/a_0	measured Δ	carrier ceiling	ν_{RAR}	measured/carrier
1.29	0.816	0.368	0.610	2.2
1.70	0.994	0.368	0.633	2.7
2.34	0.979	0.368	0.647	2.7
3.27	1.190	0.368	0.641	3.2
4.44	0.750	0.368	0.615	2.0

Table 2: The transition region, where the ceiling is most exposed. Medians of Δ in logarithmic bins of g_{bar} , beyond 2 kpc.

excess would require $a_0 \geq 1.14 \times 10^{-9} \text{ m s}^{-2}$, twelve times the canonical value and far outside the 20% spread between the two footings. We note that a joint profiling of a_0 and Υ is the formal comparison and is not repeated here; what this statistic adds is that the coherent part of the Υ freedom is externally constrained.

5 Test on clusters

We use the twelve X-COP clusters [8, 9], taking hydrostatic masses from the FORW reconstruction, gas masses from the published f_{gas} profiles, and stellar profiles for the seven clusters that ship one; the remaining five receive a radius-dependent stellar import measured from those seven and are excluded from every headline.

A radius-unit error had to be corrected first, and we verified it directly from the FITS headers rather than from any script: the gas profile’s **RADIUS** column carries **TUNIT1** = **R/R500**, with R_{500} given in each file’s own header (for example 1250 kpc for A644), whereas earlier analysis multiplied it by 10^3 as if it were Mpc. Since R_{500} ranges from 1054 to 1430 kpc across the sample, that misplaced every gas measurement by a different per-cluster factor.

With the radii corrected, the median excess over the seven clusters with measured stars is

$$\Delta/a_0 = 3.37, 2.94, 2.48, 2.28, 2.10, 1.91, 1.63, 1.52, 0.99, 0.69$$

at $r = 40, 50, 75, 100, 150, 200, 300, 420, 750, 1000$ kpc. At 40 kpc that is 9.1 times the exponential carrier’s exact bound a_0/e and 3.4 times the widest bound of any kernel in Table 1. *No choice of interpolation function can absorb this*, which is the point of proving the bound before measuring.

Both escapes close quantitatively. Restoring the ceiling by nonthermal pressure alone, with thermal and nonthermal components of similar logarithmic slope and $kT = 5 \text{ keV}$, requires a one-dimensional turbulent velocity of 1155 km s^{-1} at 40 kpc falling to 673 km s^{-1} at 300 kpc, against the Hitomi measurement of $164 \pm 10 \text{ km s}^{-1}$ in the Perseus core [10]: a factor 4.1 in velocity and 17 in pressure at the most favourable radius. Restoring it with unseen baryons requires 3.9 to 5.2 times the observed baryons inside 300 kpc, where the X-ray gas is directly imaged and dominant.

That clusters resist MOND-class gravity has been known since [11], and neutrino or other unseen components have been proposed to absorb it [12]; what the bound adds is that the requirement is exact and kernel-independent. Because the bound is a theorem, this conclusion does not depend on a fit, on a choice of interpolation function, or on a dark-matter model.

6 What the cluster residual must be, and a null

Applying the kernel to the total source, the required source mass runs $M_{\text{src}}/M_{\text{b}} = 6.5$ at 40 kpc, 8.2 at 100 kpc and 5.2 at 750 kpc, a logarithmic slope of -0.05 : it tracks the baryons. That looks like a strong constraint until the control is run. The published NFW [7] fits to the same clusters give $M_{\text{NFW}}/M_{\text{b}}$ with slope $+0.03$, and the Newtonian hydrostatic ratio $M_{\text{HSE}}/M_{\text{b}}$ likewise $+0.03$, both equally flat. Baryon-tracing therefore does *not* discriminate between an extra source and a dark-matter halo, and we report it as a null. A rescaled acceleration scale does not fit either: the a_0 that would reproduce each cluster point runs from 24 to 8 times the canonical value with slope -0.32 , so it is not a constant.

What survives as the discriminating statement is the ceiling itself, not the profile shape.

7 The carrier theorem is the bounded-boost theorem

The bound has a structural consequence for any theory in which the modification is carried by a matter-sourced scalar. Such a scalar obeys $\nabla \cdot [J_Y \nabla \phi] = 4\pi G\rho$, so on a sphere $J_Y(g_\phi) g_\phi = g_N$: the scalar force must be a single-valued function of g_N , and its longitudinal stiffness dg_N/dg_ϕ must be positive or the static problem is ill posed. Writing $g_\phi = a_0 \Delta(s)$ with $s = g_N/a_0$, that stiffness is exactly $1/\Delta'(s)$. Hence

a matter-sourced scalar can carry a kernel only up to the *maximum* of Δ , and must saturate at $C a_0$ above it.

This is the same statement as the theorem: Δ is bounded, so it has a maximum, so $s \mapsto \Delta(s)$ is not invertible beyond it. The saturation is therefore universal, not a defect peculiar to one kernel, and *every* kernel of the class leaves a constant residual scalar force $C a_0$ at high acceleration. We verify numerically that the stiffness changes sign at $g_N = 0.633 a_0$ for the exponential carrier and at $2.540 a_0$ for ν_{RAR} , exactly at the two maxima.

8 The kernel swap and its Solar-System cost

Section 5 selects ν_{RAR} over the exponential carrier, but Section 7 shows the swap is not free: the constant residual force is $C a_0$, and ν_{RAR} 's C is 1.76 times the carrier's. That residual must be screened in the Solar System, which in this action is done by the coherence length ξ in the $\xi^2 |\nabla_\perp V|^2$ operator. We price the swap on the same three gates used elsewhere in the programme — a quadrupole ceiling, the phantom mass inside Saturn's orbit, and a sunward ephemeris gate of the kind that constrains MOND in the Solar System [18] — at three external-field inputs and both footings, first validating the machinery by reproducing the previously published floors exactly on their own grid.

Two things follow. The kernel swap survives the Solar System: ν_{RAR} carried has a non-empty admissible window at both footings and all three field inputs. And a correction is required to the programme's own standing numbers: the floors 0.03/0.05 pc were computed with the *unsaturated* inverse partner, whereas the kernel a scalar can actually carry saturates and needs 0.07/0.10 pc. Any prediction evaluated at the old floors was evaluated at a coherence length the carried kernel does not permit. We recompute the principal one in the next section.

carried kernel	floor ξ , canonical	floor ξ , alt
exponential inverse partner (unsaturated)	0.03 pc	0.05 pc
exponential carrier (saturated)	0.07 pc	0.10 pc
ν_{RAR} carried (saturated)	0.10 pc	0.15 pc

Table 3: Coherence-length floors. The swap to ν_{RAR} raises the floor by a factor 1.4–1.5 and remains admissible.

9 The wide-binary prediction, recomputed

Wide binaries at 10–30 kAU probe accelerations below a_0 and have become a contested test of gravity in that regime [16, 17]; this programme carries a frozen, pre-registered estimator for the velocity ratio γ_v and a registered expectation band of 1.16–1.23 for standard MOND. Its standing prediction, $\gamma_v = 1.032$ (canonical) and 1.040 (alt), carried *two* defects that the preceding sections expose. It was evaluated at $\xi = 0.03/0.05$ pc, the floors of the unsaturated partner. And it solved with $\mu = 1 - e^{-y}$ throughout, although the wide-binary external field sits at $y_e = 1.9$, *above* the saturation point $y = 1$, where the carried interpolation is $\mu = 1 - 1/(ey)$ instead — a 5.5% difference in ν at that field.

We correct both, using the programme’s own three-dimensional pair solver and its frozen estimator, which is imported and not edited. As a control, the estimator code used here reproduces the published 1.032/1.040 to within 0.0005 when fed the original tables, so every change below is physics rather than re-implementation.

carried kernel / footing	ξ (pc)	γ_v	stale value	registered band floor
exponential carrier / canonical	0.07	1.0375 ± 0.0037	1.032	1.1614
exponential carrier / alt	0.10	1.0275 ± 0.0037	1.040	1.1917
ν_{RAR} carried / canonical	0.10	1.0450 ± 0.0037	1.032	1.1614
ν_{RAR} carried / alt	0.15	1.0300 ± 0.0037	1.040	1.1917

Table 4: The velocity ratio the pre-registered estimator returns if the candidate is true, at the corrected floors with the kernel a matter-sourced scalar can actually carry.

The change is not monotone in ξ , and it would have been wrong to assume it was. A dedicated decomposition, running the carried kernel at the *old* floors, separates the two effects: at fixed ξ the carried kernel *raises* the boost by +0.0405 (canonical) and +0.0225 (alt), because ν is larger at y_e , while raising the floor *lowers* it by −0.0350 at both footings. The two are of comparable size and opposite sign.

Two consequences. Every recomputed value lies roughly 33σ of the estimator’s own error below the registered band, so the prediction remains sharply distinguishable: the candidate returns a near-Newtonian answer where standard MOND expects 1.16–1.23. But the companion prediction weakens. The anisotropy sign-flip radius scales as $s_\times = 2.49 \xi + 0.31$ kAU, so it moves from 15.7 and 26.0 kAU at the old floors — inside the usable 4–30 kAU range — to 36–77 kAU at the corrected ones, at or beyond the edge of it.

10 The full gate ladder for the swap

Selecting a kernel on one test is not the same as adopting it. We therefore carry ν_{RAR} through the framework’s whole constraint ladder. The exercise is tractable because the carried kernel enters every gate through a single function, the coefficient of the scalar’s gradient term,

$$J_Y(s) = \frac{s}{\Delta(s)}, \quad s = g_N/a_0, \quad (4)$$

which above each saturation point grows linearly with slope $1/C$: $e = 2.718$ for the exponential carrier and $1/0.6476 = 1.544$ for ν_{RAR} . That single ratio, 1.76, is the whole content of the swap.

gate	exponential carrier	ν_{RAR} carried
PPN α_1 (bound 10^{-4})	-4.1×10^{-5}	-4.1×10^{-5}
Solar-System floor ξ , canonical / alt	0.07 / 0.10 pc	0.10 / 0.15 pc
ceiling violated, bulgeless SPARC	7.3% / 3.5%	1.2% / 0.6%
wide-binary γ_v , canonical / alt	1.0375 / 1.0275	1.0450 / 1.0300
dark-sector $ K_2 $ window	$[5 \times 10^4, 3.2 \times 10^5]$	$[2.8 \times 10^4, 2.8 \times 10^5]$
cluster ceiling violated at 40 kpc	$9.1 \times$	$5.2 \times$
linear-source screening	kernel-free	identical

Table 5: The swap priced gate by gate. It is admissible on every one.

Three of these deserve comment. *Post-Newtonian gravity is free.* In this action the scalar contributes a drag $-4(2 - K_B)/[J_Y(1 + \xi^2 k^2) + 1]$ to α_1 [14, 15]; at either kernel’s own Solar-System floor the screening factor $(\xi k)^2$ at Saturn’s orbit exceeds 10^6 , so the drag falls below 2% of α_1 and the total is $-4c_{14} = -4.1 \times 10^{-5}$ for both, an order of magnitude inside the bound. *The cosmology is free.* Re-deriving the linear-source screening from the action, its coefficient depends only on K_B , c_2 and K_2 ; the kernel’s J_Y at zero field does not appear, so the closure relation between c_2 and $|K_2|$ and the upper bound that follows from it carry over unchanged. *The dark sector barely moves.* The hydrostatic length $H = c_s^2/g$ is *identical* for the two kernels in the deep-MOND regime, $0.42 c^2/(|K_2|a_0)$; only its saturated value differs, by 0.568, so the admissible $|K_2|$ window shifts by that one factor and remains non-empty.

The costs are a factor 1.4–1.5 in the coherence length and, through it, the anisotropy sign-flip radius pushed out of the usable separation range, as the previous section records. And the one thing the swap does *not* do is the thing the theorem already forbade: clusters violate ν_{RAR} ’s wider ceiling by $5.2 \times$ at 40 kpc against $9.1 \times$ the carrier’s. Better, and nowhere near enough. That is the point of proving the bound before measuring — the cluster requirement is a property of the class, not of a choice within it.

11 What is not claimed

The theorem is exact; everything else here is a measurement with stated systematics. Hydrostatic equilibrium is assumed for the cluster masses and is tested rather than granted in the nonthermal-support calculation, but the FORW reconstruction, the distance scale and the sphericity of these systems are inherited. The SPARC test uses fixed mass-to-light ratios; a per-galaxy stellar population model would move individual points, though the Newtonian-limit calibration bounds the coherent part of that freedom. The gate ladder of Section 11 is this programme’s own constraint suite,

evaluated with its own solvers and proxies, not an independent audit. This work does not derive a_0 , does not derive $\kappa = \frac{1}{2}$, does not present a completed relativistic theory, and does not claim that any data set favours this framework over Λ CDM. Its two positive results are that the ceiling is a genuine parameter-free prediction of the modified-gravity class that a halo model cannot reproduce without fine-tuning, and that cluster cores violate it by a margin no interpolation function can absorb.

12 Reproducibility

Every number above is produced by a single committed script, `g03u_bounded_boost_theorem.py`, in the closure programme of the de Sitter–MOND repository, which carries nine checks that can fail and reports zero failures at the stated thresholds. The symbolic part of the theorem is verified in SymPy; the kernel suprema are computed on a fixed grid; the SPARC and X-COP inputs are the public data products, read with their own headers.

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